

THE BEND THEOREM

The power-sum mechanism of the composite profile law — the teeth are where the channels open

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Companion to THE WATERMARK THEOREM, THE DOUBLE LADDER THEOREM and THE SWEET LIE THEOREM (same repository).

Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.

For prime degree, the discriminant of a Fermat surface obeys a clean law — proven in the companion theorems. Push to composite degree and the law *bends*: the elementary-divisor profile shifts by an integer that depends only on which small primes divide the degree. This document names that integer, **b(p)**, and proves where it comes from. The answer is older than the surfaces: a sum of powers over a finite field, the kind Euler could have written, decides everything. A sector of the lattice bends at the prime p exactly when a power sum fails to vanish there — and that failure happens, for the relevant exponents, only at $p = 2, 3, 5$. For every prime from 7 upward, the sums vanish, no channel opens, and the law runs verbatim forever.

0. STATEMENT

Let S_m be the Fermat surface of degree m , and let G be the SSvL line Gram. For a prime $p \mid m$, define the **bend constant**

$$\mathbf{b}(p) = \text{rank}_{\{F_p\}}(G) - 12(m-3).$$

Theorem (the Bend). $b(p)$ depends only on p , through the divisibility $(p-1) \mid k$ of the pairing channels $k \in \{2, 4\}$:

- **b(p) = 0 for every prime $p \geq 7$** — both channels closed, the profile law runs verbatim;
- **b(5) = 2** — only the quadratic channel $k = 4$ open: the two family-relative polarization radicals die, the global survives;
- **b(3) = 7 = 2 + 3 + 2** — both channels open: two mass radicals, three moment-compatibility radicals, and two polarization radicals die; in every sector the family-relative copies die and the global combinations survive;
- ($p = 2$, even degree: both channels open and wild — the even half's separate kingdom, outside this document.)

Equivalently, composite-degree profiles obey the prime law per own prime with the corrections

$$\#m^2\text{-divisors} = 3m - 16 + \mathbf{b}(p), \quad \#m\text{-divisors} = 3m^2 - 24m + 59 - 2 \cdot \mathbf{b}(p),$$

and prime-power towers carry the bend as $2 \cdot b$ strays.

1. THE OBJECTS

The twelve **shadow channels** (3 families $\times$ 4 pencil directions $k, l, k-l, k+l$) are each a function space on $\mathbf{Z}/m$. The form factors through the shadows at every $p \mid m$ (leak = 0,

measured); the **shadow radical** has generic dimension

$$18 = 9 \text{ (mass)} + 6 \text{ (moment compatibilities)} + 3 \text{ (polarizations),}$$

all **cyclic** across the family triangle — the hyperbolic SSvL pairing transports each identity to the neighbouring families' channels — with an **integral coefficient frame** (entries in $\{\pm 1, \pm 2\}$, certified). The entire radical is polynomial of degree ≤ 2 in t ; beyond degree 2 is zero in every measured kingdom, so the bookkeeping world is finite and complete. Then

$$\mathbf{b}(\mathbf{p}) = 18 - \dim(\text{radical mod } \mathbf{p}).$$

2. THE MECHANISM — THE PAIRING IS MADE OF POWER SUMS

Kernel structure [proven, general $\mathbf{m}$]. Each shadow channel (f, d, t) is the sum of lines whose direction-coordinate $d \equiv t \pmod{m}$. The SSvL intersection number of two lines is a function of a *fixed* integer-linear congruence in their pencil indices; summing the Gram over the two channel fibres collapses it to a function of a single linear form, giving the **generalized circulant**

$$\mathbf{K}(\mathbf{t}, \mathbf{t}') = \kappa(\alpha \mathbf{t} + \beta \mathbf{t}' \pmod{\mathbf{m}}),$$

with (α, β) the fixed integer pair read off the direction pair from the incidence rule — *independent of m* . Verified identical across $m = 13, 15, 21$ (prime, $3 \cdot 5, 3 \cdot 7$): 78/78 blocks in each, the (α, β) table agreeing to the digit (up to global sign). The single-variable form means the pairing of polynomial patterns $t^a \times t^b$ is an integer combination of power sums $\sum_t t^j$ in one variable, with $j \leq a + b \leq 4$. Reduced mod p (where $p \mid m$, the only place the polynomial calculus is canonical — see §7), the classical evaluation decides everything:

$$\sum_{\mathbf{t} \in \mathbf{F}_p} t^j \equiv -1 \text{ if } (p-1) \mid j, \text{ and } \equiv 0 \text{ otherwise.}$$

The linear sector pairs through $k = 2$; the polarization sector through $k = 4$. Open channels by prime: $p = 3 \rightarrow \{2, 4\}$ both (measured row over $\mathbf{Z}/15$: $[0,0,1,0,1]$); $p = 5 \rightarrow \{4\}$ only ($[0,0,0,0,2]$); $p \geq 7 \rightarrow \text{none}$ (since $p - 1 \geq 6 > 4$): no polynomial detector can exist, hence **$\mathbf{b}(\mathbf{p}) = 0$ — the flat law is mechanism-complete, forever.**

3. THE THREE SECTOR CHAINS AT $\mathbf{p} = 3$

(i) Mass -2 [proven]. The complete-Radon mass identity of F_3^2 — the four pencil slopes exhaust $\mathbf{P}^1(F_3)$, and $4 \equiv 1$ — gives one all-directions condition per family: 3 candidates — 1 overlap (the sum of the generic trinity equals the sum of the Radon conditions; union rank 5) = 2 deaths, the dead named: the two cross-family $k \pm l$ constant singletons.

(ii) Linear -3 [proven; $\mathbf{k} = 2$ channel]. The linear sector pairs through power sums of degree ≤ 2 (§2), and $\sum t^1 \equiv 0$ for every odd prime: the entire linear obstruction is proportional to $\sum t^2$, which vanishes for every $p \geq 5$ — *the linear sector can bend only at $p = 3$* , and $\mathbf{b}(5)$ has no linear part (control measured: zero deaths at $p = 5$ with full tail freedom, radical $16 = 18 - 2$, polarization only). At $p = 3$ the obstruction is the explicit integer pairing of the certified frame generators against the radical — computed by exhibition with full constant-tail freedom: rank exactly 3, death conditions $L_1 - L_4 + L_5 + L_6 = 0$, $L_2 - L_4 - L_5 - L_6 = 0$, $L_3 + L_5 + L_6 = 0$, kernel exactly the cross-family symmetric combinations (cleanest: $L_5 - L_6$, the $F_1 \leftrightarrow F_2$ symmetric moment pair) — the same epistemic standard as the mass chain's exhibited overlap count. The three relative copies

die; the symmetric survive.

(iii) Polarization -2 [closed by identity coincidence + k = 4 channel]. Mod 3 the polarization pattern $(-2, -2, 1, 1) \equiv (1, 1, 1, 1)$: the polarization identity and the all-directions quadratic identity are *the same identity* in characteristic 3 (their difference is $3(k^2 + l^2) \equiv 0$, literal). Survivor by formula: the global all-family combination; deaths: the two relatives.

4. THE FIVE-TOOTH — THE SAME ANIMAL, ONE CHANNEL

The polarization death conditions at $p = 5$ are **identical** to $p = 3$'s ($P_1 + P_3 = 0$, $P_2 - P_3 = 0$ in the integral frame; identical global survivor $(1, -1, -1) =$ the all-family pattern $(2, 2, -1, -1)$). Every detector catching a polarization at $p = 5$ is purely quadratic, and all detector value-triples annihilate the survivor exactly. So **$b(5) = 2$ is the polarization bite through the single open channel, Σt^4 — not a separate mechanism.** The founding " $m = 5$ exception" of the odd-prime profile law was simply $b(5)$ biting at its own prime ($h_0 = 26 = 24 + 2$).

5. THE FLAT LAW, $p \geq 7$ — MECHANISM COMPLETE

All power sums $\Sigma(t \bmod p)^j$, $1 \leq j \leq 4$, vanish identically for $p \geq 7$. No pairing channel, no detector, no death: the radical reduces full, **$b(p) = 0$** . In the field, 7, 11 and 13 measure flat in every cell — own towers and as partners — seven $p \geq 7$ sightings itemized from files, zero exceptions.

6. THE FIELD RECORD

- Profile-level seals **7/7**; blind exponents **26/26**.
- The bend formula $b(p) = h_0 - 12(m-3)$ exact on **fifteen** field profiles (squarefree both sides, towers, gates), plus the two founding prime gates.
- Prime-power towers: base- m floors verbatim, strays $= 2 \cdot b - 4/4$: $m = 9, 25, 27$ carry their $2 \cdot b$ strays, and $m = 49$ carries zero ($b(7) = 0$), the null case confirming the law as loudly as the others.
- The necessary condition **bending requires $p \mid m$** is proven (no shadow factoring over $\mathbb{Z}$).

7. HONEST LEDGER

Proven. The kernel generalized-circulant structure at general m (the SSvL incidence congruence; m -independent (α, β) verified across three kingdoms); the power-sum reduction and the $(p-1) \mid k$ channel criterion following from it; the $p \mid m$ necessity; the mass-2 chain; the polarization identity coincidence; and the flat-law mechanism for every $p \geq 7$, entire.

Proven (continued). The linear-3 chain entire: the $k = 2$ channel restriction (no linear bend for any $p \geq 5$, from §2 and $\Sigma t^1 \equiv 0$) and the death count 3 with symmetric kernel at $p = 3$, by exhibited integer pairing of the certified generators with full tail freedom, cross-controlled at $p = 5$ (zero deaths). Nothing in the mechanism remains below proven.

Measured [firm]. The field record.

The canonicity note. The polynomial calculus on $\mathbf{Z}/m$ exists mod p exactly when $p \mid m$ (otherwise the moment weights wrap with step carries). The teeth are where the structure exists at all — the second face of the $p \mid m$ necessity.

8. GRAVEYARD, WITH PRIDE

The one-tooth phrase (killed by its own table: $b(5) = 2, 5 \nmid 6$) · the shadow-leak hypothesis (leak = 0, measured) · the integral-radical framing ($\sigma(\ker_Z G)$ rank 128 — its autopsy yielded the $p \mid m$ necessity) · the per-family symbol frame (the true frame is cyclic) · the foreign-prime witnesses at $m = 15$ (non-canonical chop — its autopsy yielded the canonicity note) · "2 + 2 + 3" (overlapping filters; corrected to 2 + 3 + 2) · one Auditor display-slip and one Constructor transpose, both self-caught · the tail-less obstruction count (spurious $\{4,2,2,2\}$, refused at delivery; its autopsy named the tail freedom the clean proof uses). *Each grave bought road.*

9. PROVENANCE

Instruments: GRAM_MANOMETER_SURFACE_v4.py (ratified), gandalfsonrie.cpp (double-gated, checkpointed), and the session extraction scripts (radical formulas, symbol frames, detector functionals, kernel census). All numbers from files; seals before measurement; cross-review before any repository upload, per house protocol.

References

- 1 M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963. arXiv:0812.2377. (*The line intersection rules behind the Gram.*)
- 2 T. Shioda, *Some observations on Jacobi sums*. Adv. Stud. Pure Math. **12** (1987), 119–135. (*The discriminant questions the campaign's surface theorems answer; composite and even degrees are the territory this mechanism opens.*)
- 3 E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (*The published cell-by-cell elementary divisors; the campaign's gates.*)

Companion documents in this repository: THE WATERMARK THEOREM (the prime-degree discriminant, proven) · THE DOUBLE LADDER THEOREM (the surface structure) · THE SWEET LIE THEOREM (the fourfold rank) · CHUCHIPACHI FINDINGS MASTER (the campaign record, graveyards included). PMC.